Video Encoding for Storage and Quality: A Multi-Provider Benchmark

Comparing compression efficiency across cloud encoding providers for e-commerce, ugc, archives, security & surveillance workloads

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About This Report

This report was commissioned by Vidaio and produced by Nikola Rahman, an independent video engineering consultant. All methodology, test content, analysis, and conclusions are the author's own.

- All providers were tested using publicly available APIs with their best available quality settings.
- Provider-specific configuration details are documented in the Methodology section.
- Draft results were shared with all tested providers prior to publication for factual accuracy review.

Providers considered but excluded: Mux and Cloudflare Stream were initially evaluated but excluded from the final benchmark - both produce H.264-only static MP4 downloads with no quality controls. AV1 and H.265 are only available via streaming, not as downloadable files, making fair comparison with the other providers impossible.

1. Executive Summary

The gap

Existing video encoding benchmarks evaluate providers through a streaming lens - adaptive bitrate ladders, segment-based delivery, and startup latency. No benchmark exists for **quality-optimized**

single-file encoding: the use case that matters for archival storage, e-commerce product videos, UGC platforms, and online education.

What we tested

30 video clips across four industry verticals, encoded through 5 providers using AV1 and H.265, measured on VMAF perceptual quality, file size, and compression efficiency.

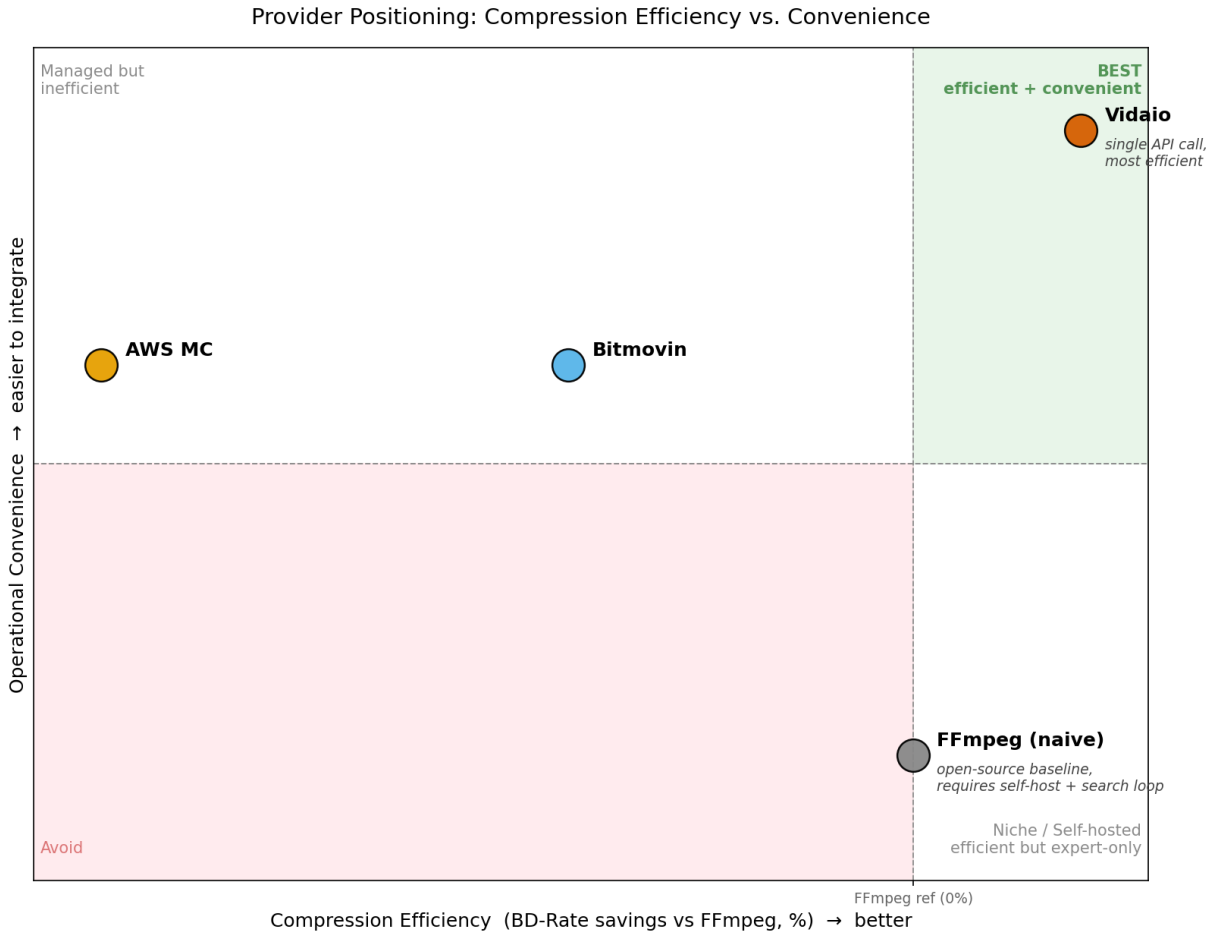
Headline findings

Three charts summarise the benchmark: where each provider sits on the two dimensions that drive a buying decision (Figure 1.1), how many bits each provider uses to deliver each VMAF target (Figure 1.2), and how often each provider strictly meets that target (Figure 1.3). Per-codec breakdowns and BD-Rate tables are in §4.

Beyond compression, Vidaio differs in how it works: it hits a VMAF target in a single encode, while every other provider runs an iterative search of several encodes per target. That single-pass workflow is the operational edge in Figure 1.1, and it carries through to lower per-delivered-minute cost (§6) and faster turnaround (§7).

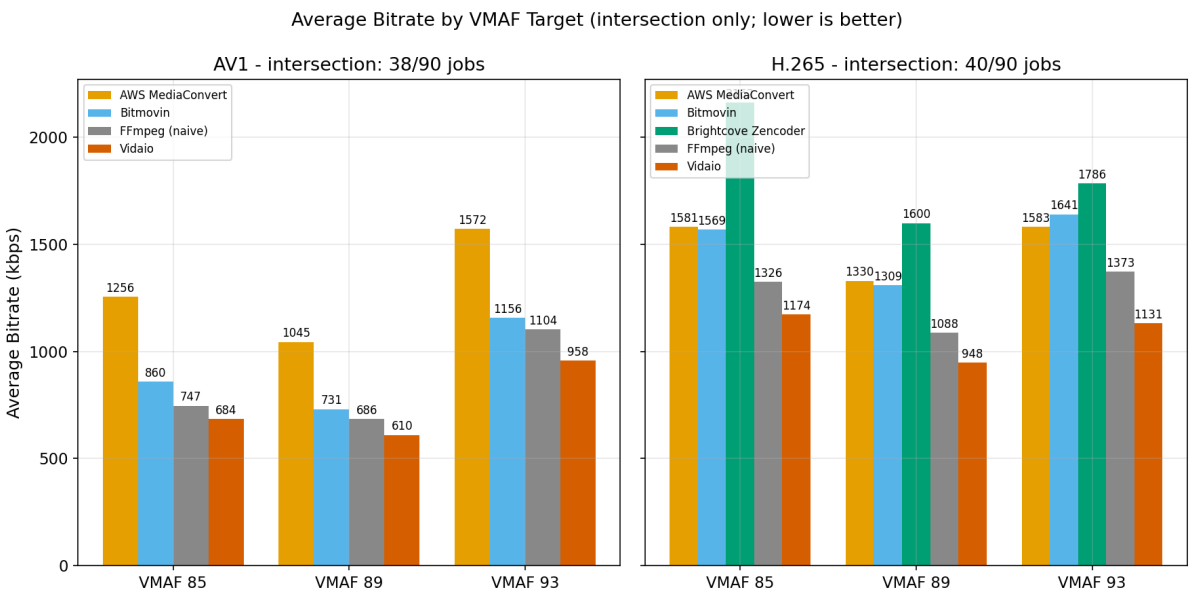
On grainy archive film, only Vidaio and self-hosted FFmpeg reach the quality targets at all — the other managed services (Bitmovin, AWS MediaConvert, Brightcove Zencoder) miss by a wide margin, a limitation visible throughout §4 and §5.

Figure 1.1 - Provider Positioning: Compression Efficiency vs. Convenience



X-axis (Compression Efficiency): BD-Rate savings vs the FFmpeg reference, averaged across both codecs — positive means the provider uses fewer bits than FFmpeg at matched quality. Y-axis (Operational Convenience): score from the §1.4 capability matrix (native VMAF targeting, single-encode workflow, managed service, low codec-parameter expertise). Brightcove Zencoder is omitted — H.265 only, not comparable across both codecs.

Figure 1.2 - Average Bitrate by VMAF Target



Grouped bars per provider at each of the three VMAF targets. Lower is better — smaller bars mean fewer bits to deliver the same quality.

Operational Capabilities

Capability	Vidaio	Bitmovin	AWS MC	Zencoder	FFmpeg (naive)
Targets VMAF natively (API parameter)	✓	—	—	—	—
Single-encode workflow (no search loop)	✓	—	—	—	—
Managed service (no self-hosting)	✓	✓	✓	✓	—
No codec-parameter expertise required	✓	partial	partial	partial	—

Vidaio is the only encoder that accepts a target VMAF as a first-class API input. Every other encoder requires the user to wrap it in an iterative search loop, configure rate-control/bitrate/profile parameters, and run external VMAF measurement between iterations. FFmpeg additionally requires self-hosting the encoder.

2. Why This Benchmark Exists

The streaming assumption

All major cloud encoding benchmarks - Ozer (2022), MSU (2020, 2022) - evaluate providers through a streaming lens: ABR ladders, segment-based delivery, per-title optimization for adaptive playback. This methodology makes sense for OTT delivery, but it ignores customers whose goal is the **smallest possible file at a target quality**, not adaptive playback.

Who needs single-file encoding

Segment	Use case	Why streaming benchmarks don't apply
E-commerce	Product videos as progressive downloads	Millions of short clips; ABR overhead exceeds the video itself
UGC platforms	Ingest, transcode, store uploaded video	Storage cost is the dominant concern, not adaptive playback
Video archives	Long-term preservation at minimum cost	Every byte saved compounds over years and petabytes
Online education	Lecture and tutorial content at scale	Text-heavy content with consistent quality requirements
Security & surveillance	24/7 recording with months of retention	Massive volume; storage cost dominates; low motion with rare events

How streaming-optimized encoding differs from single-file encoding

Streaming encoders typically use short GOPs (2-4 seconds) to enable segment-based switching. Each GOP starts with an I-frame - a full-frame image that costs 10-50x more bits than a P-frame. For single-file delivery where segment switching is not needed, these additional I-frames consume bits without a corresponding benefit to the viewer.

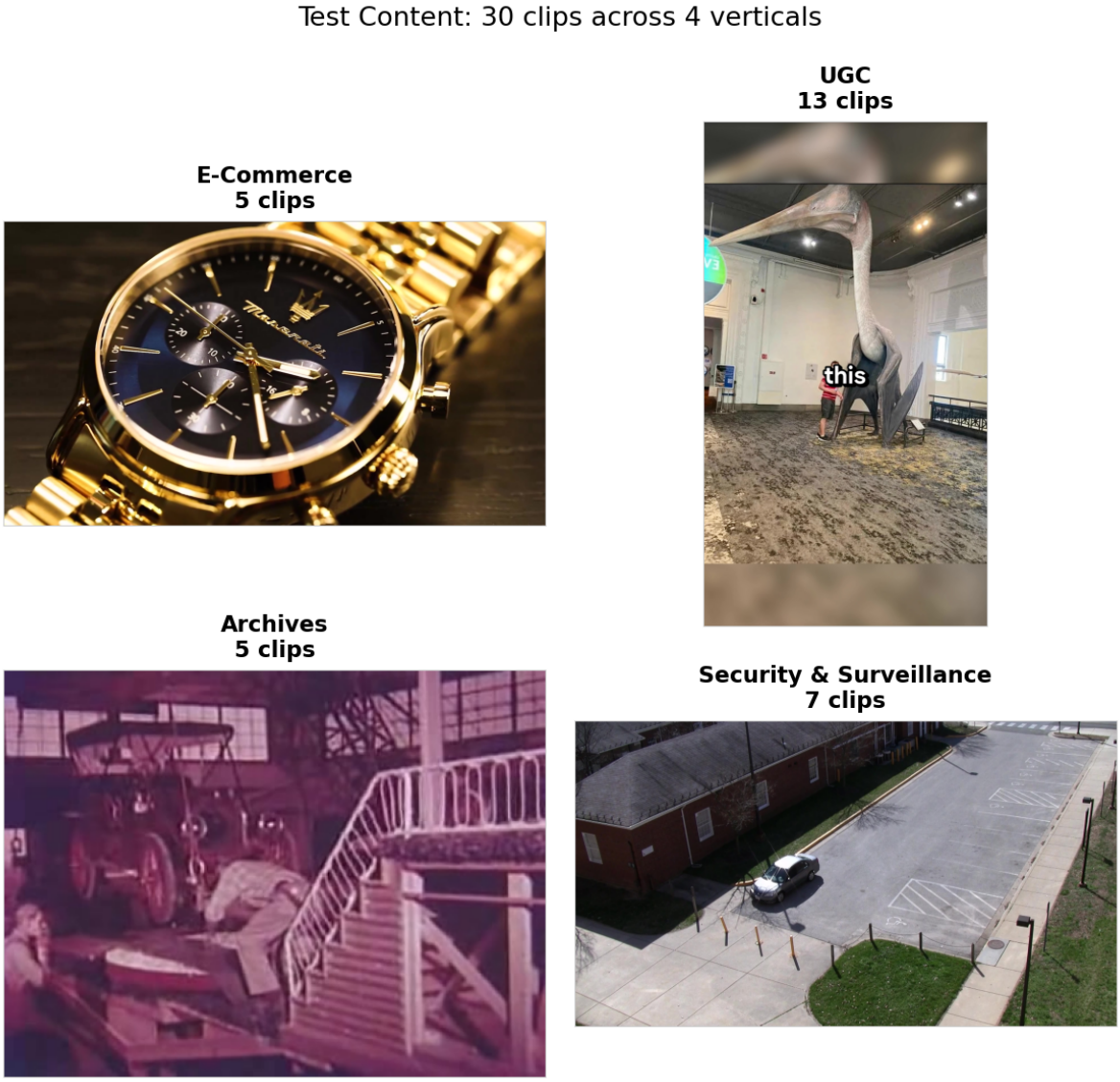
3. Methodology

3.1 Test Content

30 source videos across the four verticals, chosen to span a diverse range of spatial and temporal complexity. Synthetic test sequences (ITU reference clips, Blender demos) were avoided in favor of content representative of real production workloads. The content gallery and SI/TI scatter below show the corpus; per-clip provenance is in Appendix B.

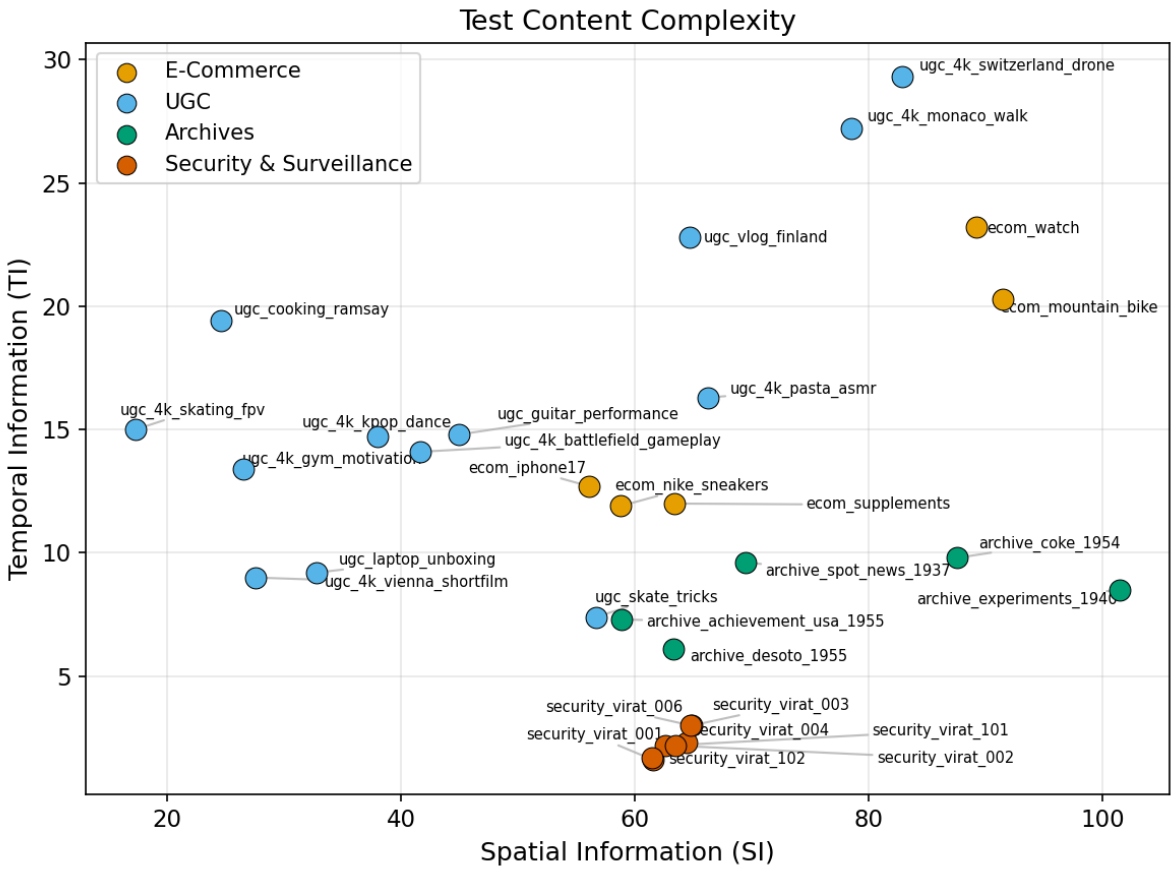
Input preparation: All source files were re-encoded to H.264 CRF 18, no audio, MP4 container. Videos longer than 120 seconds were trimmed from the start. Pixel aspect ratio is normalized to 1:1 (square pixels), so encoders that auto-rescale non-square sources produce outputs at the same dimensions as the reference and VMAF can be measured directly.

Figure 3.1 - Test Content Gallery



Representative frame from each test clip, organized by vertical.

Figure 3.2 - Spatial-Temporal Complexity Scatter



Each point is one test clip, color-coded by vertical. The test corpus covers a diverse range of content complexity.

3.2 Providers and Configuration

Provider	Rate Control	Quality Parameter	Codec Support	Black-box?
AWS MediaConvert	QVBR	QvbrQualityLevel (1-10) + fine-tune	H.265, AV1 (MULTI_PASS_HQ)	No
Bitmovin	CRF	crf (0-51 for H.265, 0-63 for AV1)	H.265, AV1	No
Brightcove Zencoder	Quality	quality (1-5)	H.265 only (no AV1 support)	No
FFmpeg (naive)	CRF	crf (0-51 for libx265, 0-63 for libsvtav1)	H.265 (libx265 preset slow), AV1 (libsvtav1 preset 4)	No
Vidaio	VMAF-targeted	Target VMAF (low/medium/high → 85/89/93)	H.265, AV1	No

Fairness statement: Each provider was configured to produce the smallest possible file size at a given target quality (VMAF). For providers with quality controls, an interpolation search over each provider's quality parameter finds the setting that hits each VMAF target. For providers without quality controls, the single output represents the provider's best-effort encoding.

Configurable providers: For each clip, an interpolation search over each provider's quality parameter finds the setting that hits target VMAF scores (85, 89, 93). This produces matched-quality data points for direct comparison.

Encodes per target — the central methodological asymmetry. Every provider in this benchmark *except Vidaio* is driven through this iterative VMAF search — typically three to five encodes per target — to find the setting that lands on each score; only the best-fitting encode is reported. Vidaio accepts the target VMAF as a native API input and returns a *single encode*. This cuts both ways. Vidaio cannot iterate to clear a target exactly, so it occasionally lands just below it (almost always within 1 VMAF point — see §4.3); but it reaches that result in one encode rather than three to five, which is what drives the per-target pricing (§6) and encoding-time (§7) gaps. Reported BD-Rate and RD curves use each provider's best on-target encode, so the efficiency comparison is on delivered compression, independent of how many attempts it took.

FFmpeg (naive) baseline: In this benchmark "FFmpeg (naive)" is self-hosted FFmpeg with `libx265` (H.265, preset `slow`) and `libsvtav1` (AV1, preset 4), driven by an interpolation search over the single CRF quality knob. No content-aware preset tuning, no smart rate control — the same encoder settings are applied to every clip and only CRF varies between iterations. These are the same open-source encoders that the managed providers (Vidaio included) build their pipelines

on; the benchmark is therefore comparing the parameter-selection and search wrappers around the encoders, not the underlying codecs.

3.3 Quality Measurement

- **VMAF model:** `vmaf_v0.6.1neg`.
- **VMAF computation:** full-video, frame-by-frame.
- **Bitrate:** measured via ffprobe on the output file.
- **BD-Rate:** Bjontegaard delta rate via PCHIP interpolation over the overlapping VMAF range.

Note: The VMAF model is dictated by Vidaio's API, which only accepts `vmaf_v0.6.1neg` for quality targeting. We standardized on it across all providers for fair comparison.

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4. Results by Codec

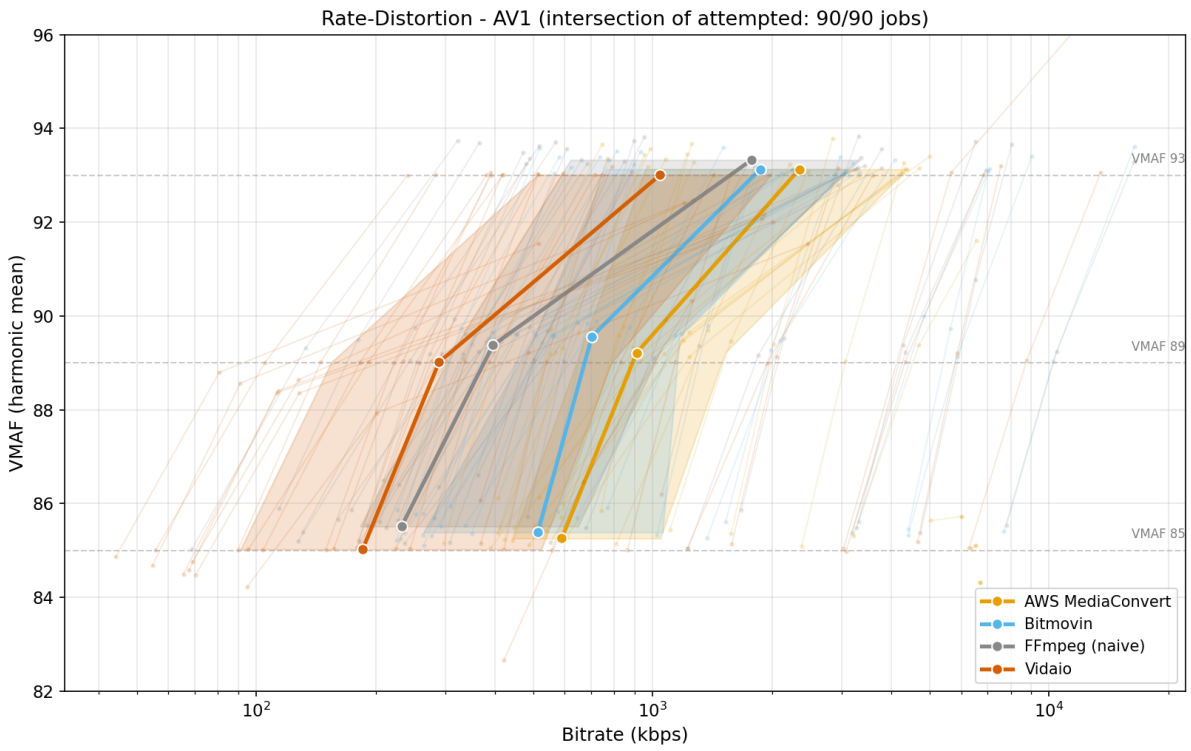
4.1 AV1 Results

BD-Rate is computed on the *intersection* — the (clip, target VMAF) jobs where **every provider** in this codec converged. Reference is **FFmpeg** (the open-source baseline; negative BD-Rate means the provider uses fewer bits than FFmpeg at matched quality). Intersection: **38 / 90 jobs**. It is smaller than the full matrix because BD-Rate requires every provider on the same jobs, and the grainy archive clips — where the managed cloud encoders top out near VMAF 36, far below target (§4.1.5) — drop out for all of them.

Provider	Avg VMAF	BD-Rate vs. FFmpeg	Rank
AWS MediaConvert	89.2	+57.6%	4
Bitmovin	89.2	+10.2%	3
Brightcove Zencoder	-	-	-
FFmpeg (naive)	89.3	ref	2
Vidaio	88.9	-7.8%	1

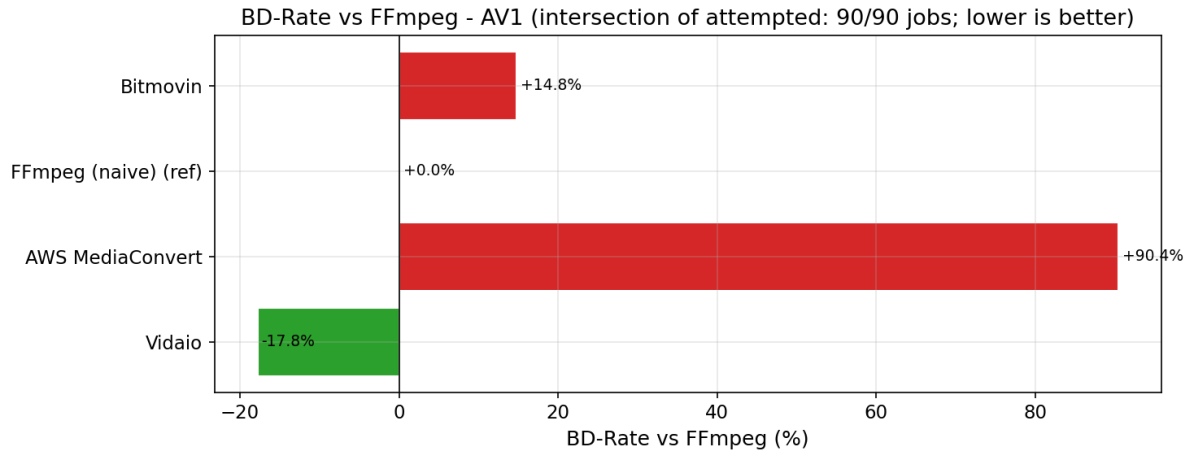
4.1.1 Aggregate AV1 Rate-Distortion

Figure 4.1.1 - Aggregate RD Curves, AV1



All providers, AV1 encoding, averaged across all test clips. Horizontal lines at VMAF targets (85, 89, 93).

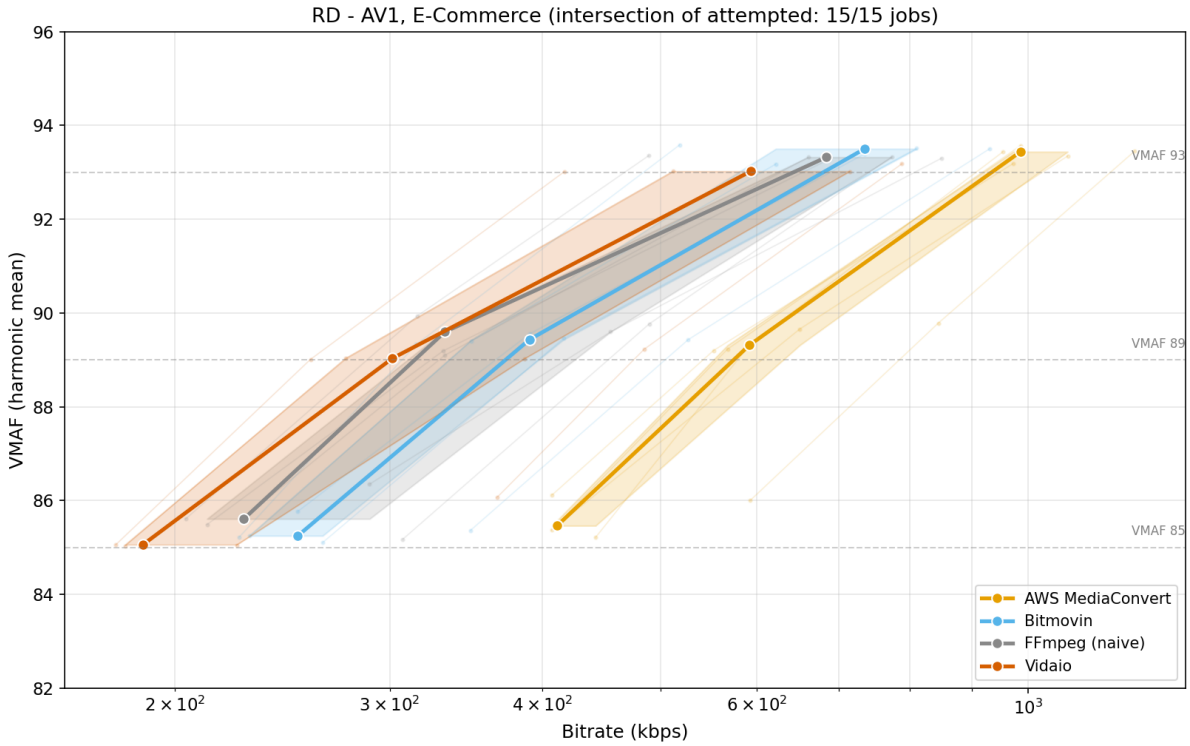
Figure 4.1.2 - BD-Rate Summary, AV1



Percentage bitrate difference relative to the most efficient provider. Negative = saves bitrate at equivalent quality.

4.1.3 AV1 Results - E-Commerce

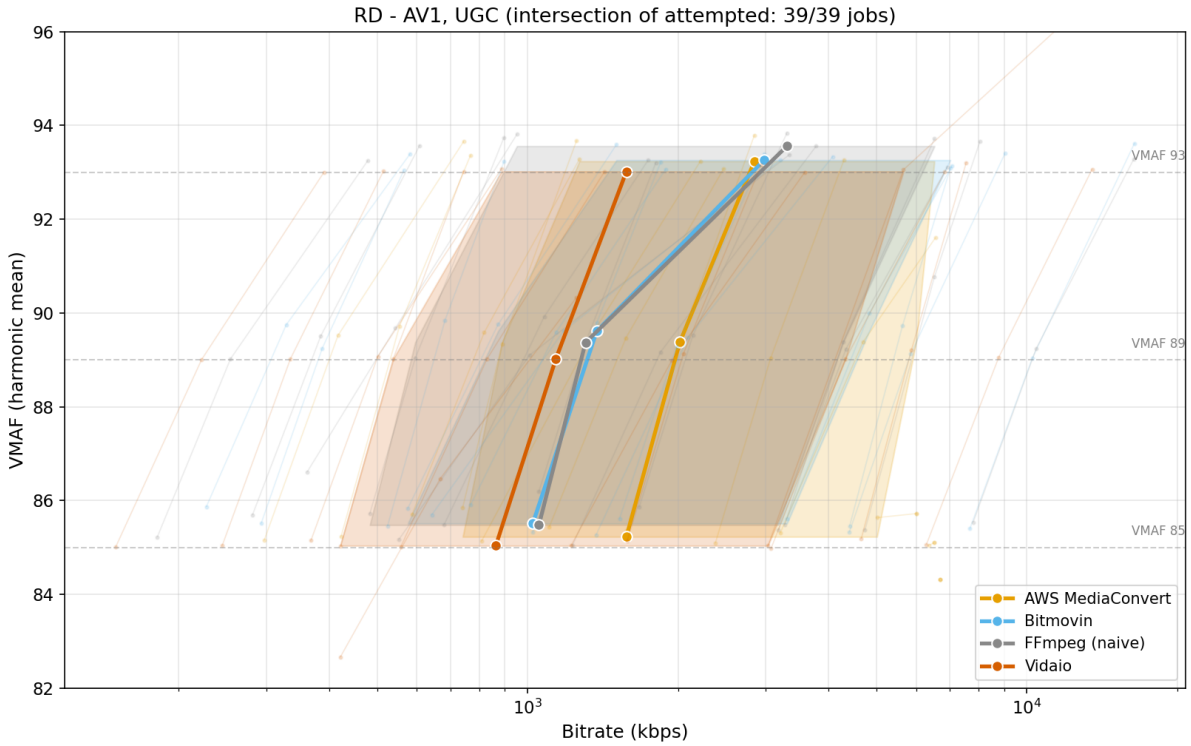
Figure 4.1.3 - RD Curves, AV1, E-Commerce Clips



Short product clips with clean studio motion; every encoder reaches the targets, so the curves separate on bitrate alone. Brightcove Zencoder encodes H.265 only and is absent from AV1 charts.

4.1.4 AV1 Results - UGC

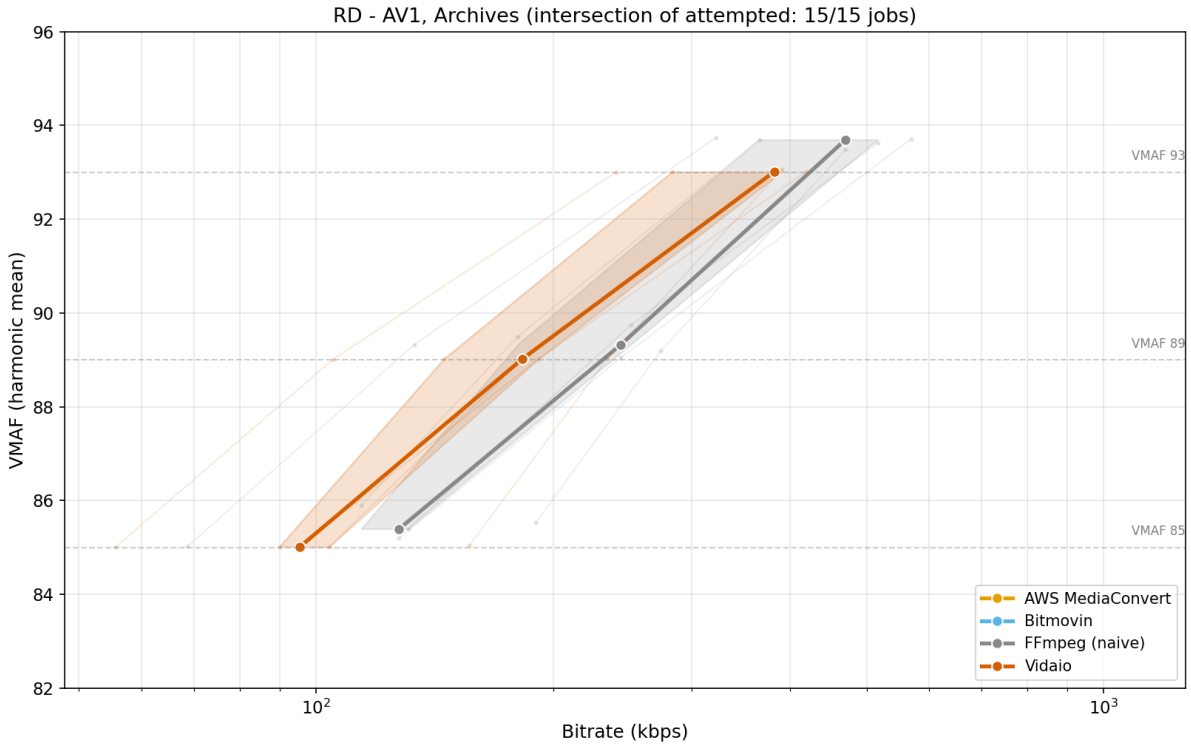
Figure 4.1.4 - RD Curves, AV1, UGC Clips



Mixed-complexity user footage including the 4K subset; the curves fan out as content difficulty rises. Brightcove Zencoder encodes H.265 only and is absent from AV1 charts.

4.1.5 AV1 Results - Archives

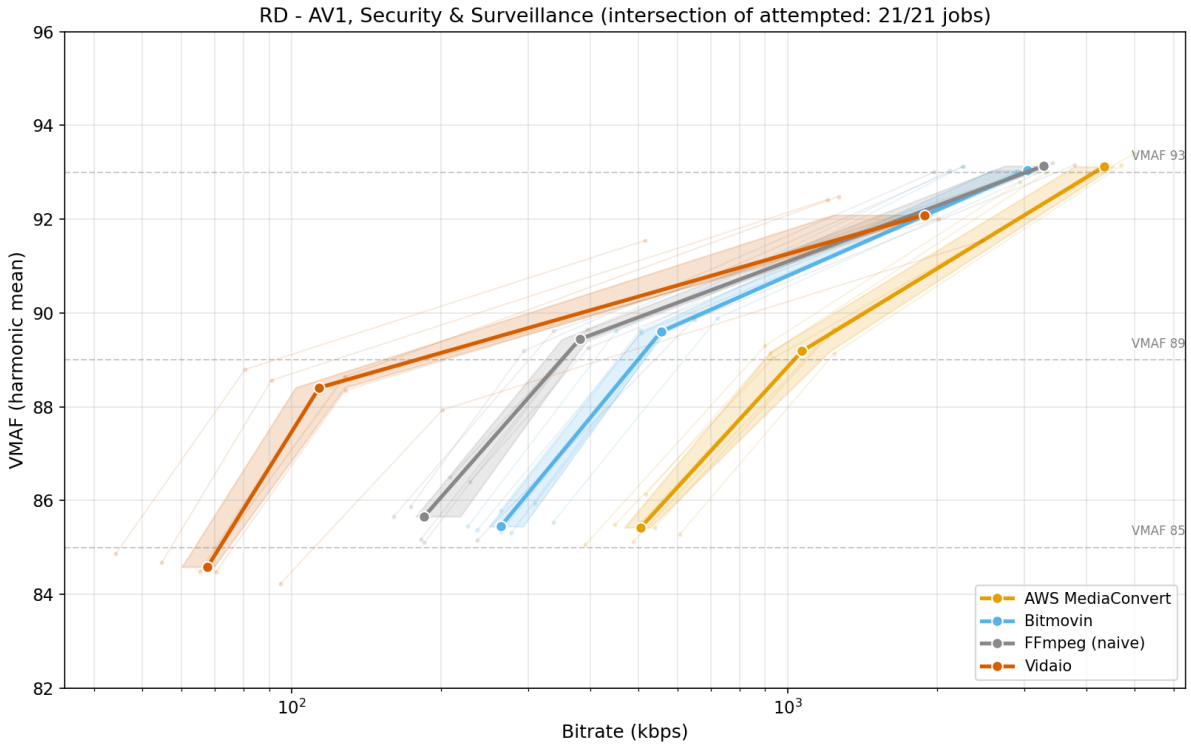
Figure 4.1.5 - RD Curves, AV1, Archives Clips



Grainy archive film: Bitmovin, AWS MediaConvert and Brightcove Zencoder scored VMAF 19-59 (avg 36) against targets of 85+ and fall below the plotted range, leaving only Vidaio and FFmpeg.

4.1.6 AV1 Results - Security & Surveillance

Figure 4.1.6 - RD Curves, AV1, Security & Surveillance Clips



Low-motion surveillance content compresses tightly and the encoders cluster closely. Brightcove Zencoder encodes H.265 only and is absent from AV1 charts.

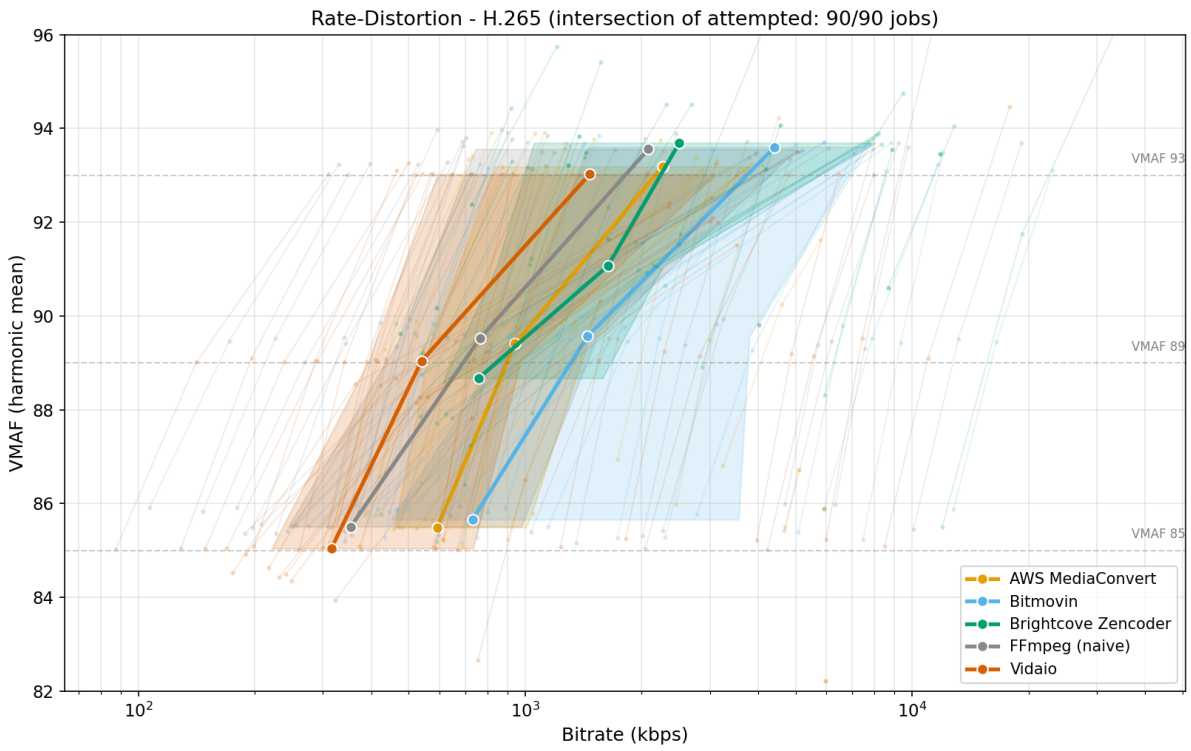
4.2 H.265 Results

BD-Rate is computed on the *intersection* — the (clip, target VMAF) jobs where **every provider** in this codec converged. Reference is **FFmpeg** (the open-source baseline; negative BD-Rate means the provider uses fewer bits than FFmpeg at matched quality). Intersection: **40 / 90 jobs**. It is smaller than the full matrix because BD-Rate requires every provider on the same jobs, and the grainy archive clips — where the managed cloud encoders top out near VMAF 36, far below target (§4.2.5) — drop out for all of them.

Provider	Avg VMAF	BD-Rate vs. FFmpeg	Rank
AWS MediaConvert	89.4	+15.1%	3
Bitmovin	89.3	+19.9%	5
Brightcove Zencoder	91.8	+18.0%	4
FFmpeg (naive)	89.3	ref	2
Vidaio	88.8	-9.7%	1

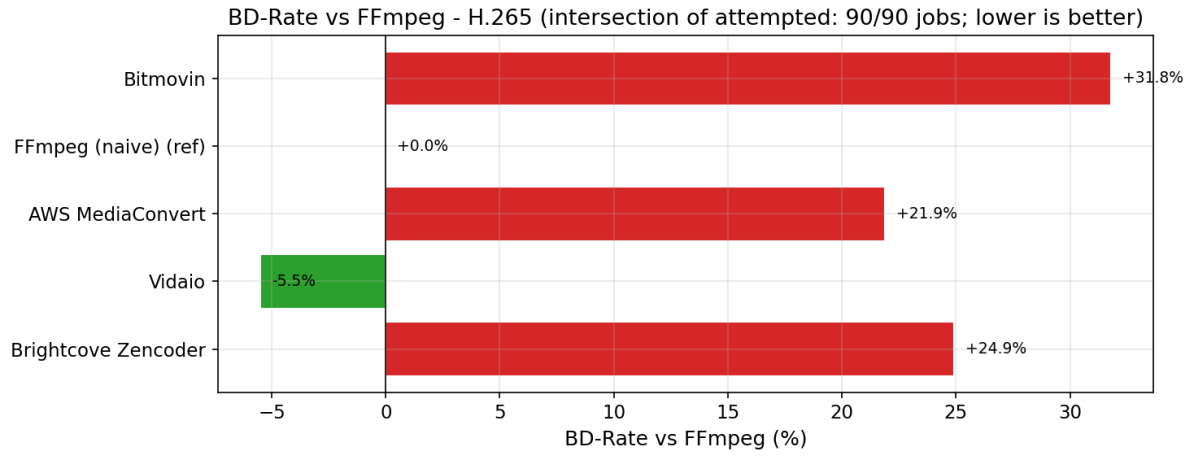
4.2.1 Aggregate H.265 Rate-Distortion

Figure 4.2.1 - Aggregate RD Curves, H.265



All providers, H.265 encoding, averaged across all test clips. Horizontal lines at VMAF targets (85, 89, 93).

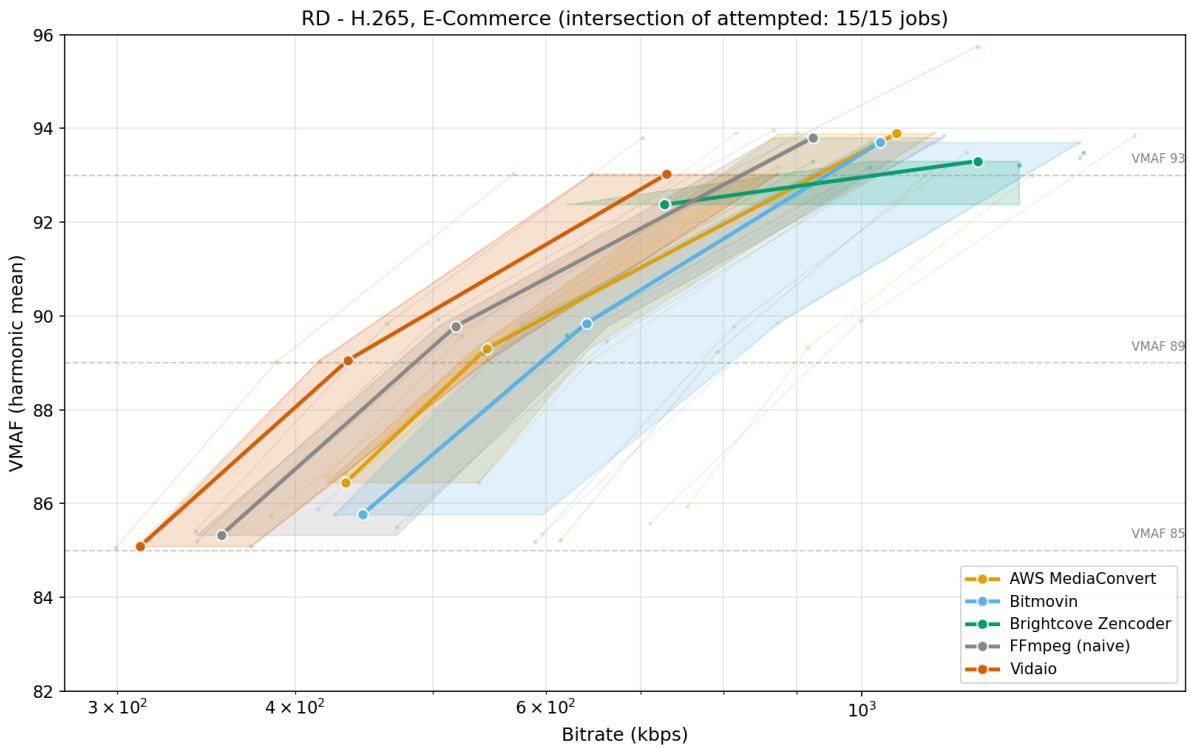
Figure 4.2.2 - BD-Rate Summary, H.265



Percentage bitrate difference relative to the most efficient provider. Negative = saves bitrate at equivalent quality.

4.2.3 H.265 Results - E-Commerce

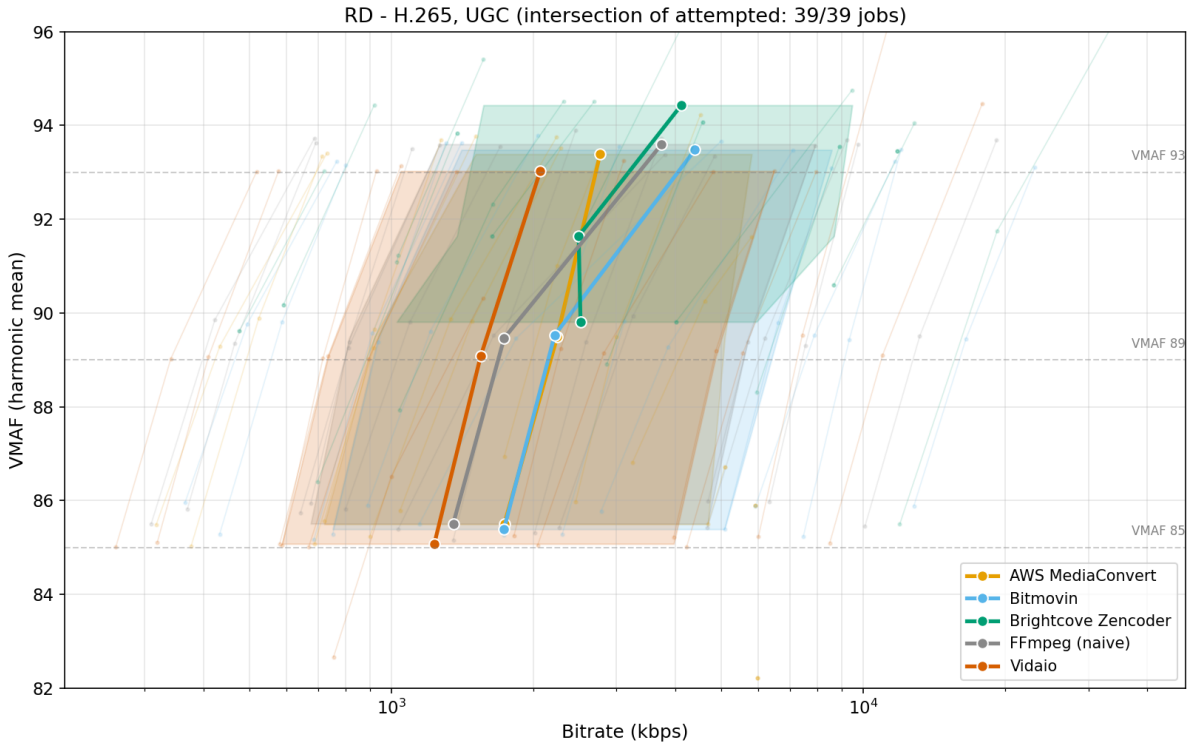
Figure 4.2.3 - RD Curves, H.265, E-Commerce Clips



Short product clips with clean studio motion; every encoder reaches the targets, so the curves separate on bitrate alone.

4.2.4 H.265 Results - UGC

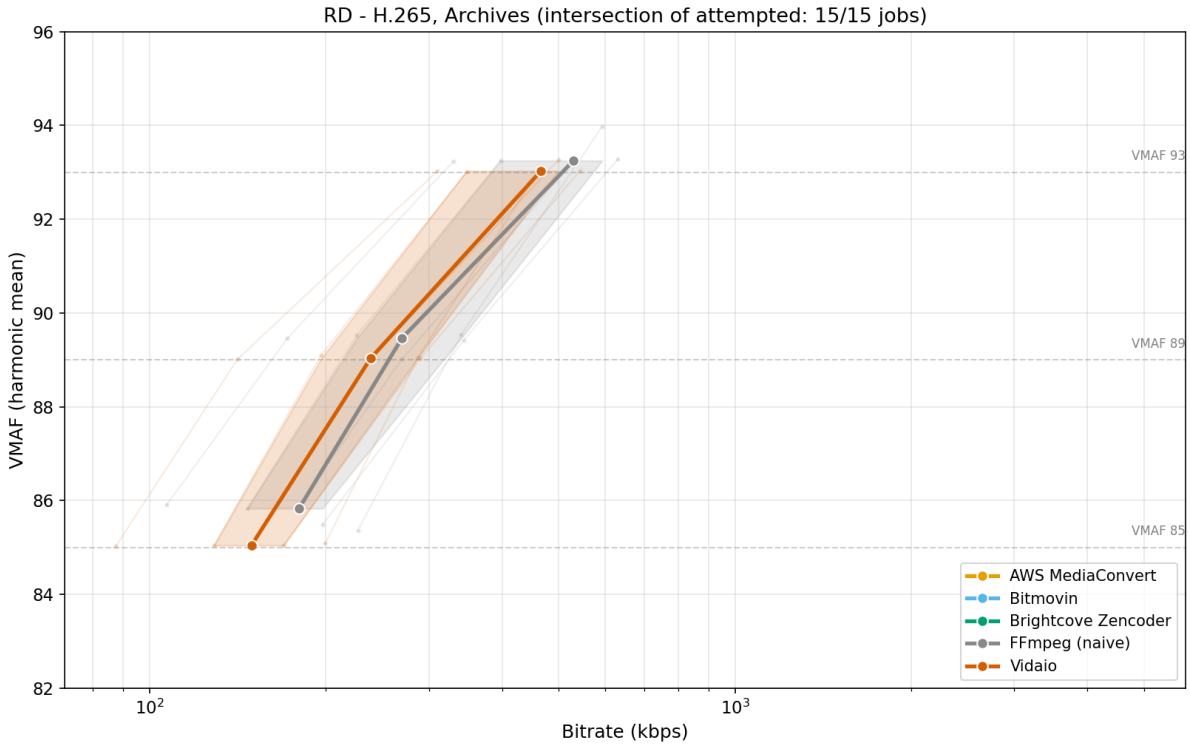
Figure 4.2.4 - RD Curves, H.265, UGC Clips



Mixed-complexity user footage including the 4K subset; the curves fan out as content difficulty rises.

4.2.5 H.265 Results - Archives

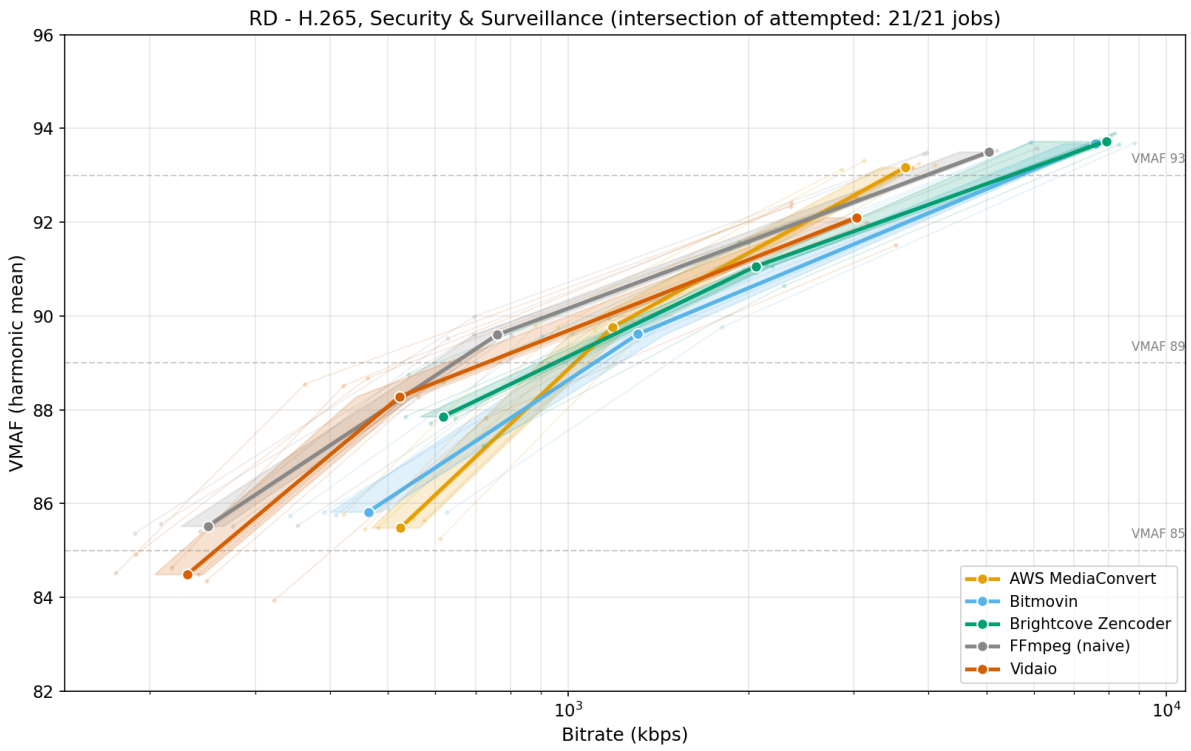
Figure 4.2.5 - RD Curves, H.265, Archives Clips



Grainy archive film: Bitmovin, AWS MediaConvert and Brightcove Zencoder scored VMAF 19-59 (avg 36) against targets of 85+ and fall below the plotted range, leaving only Vidaio and FFmpeg.

4.2.6 H.265 Results - Security & Surveillance

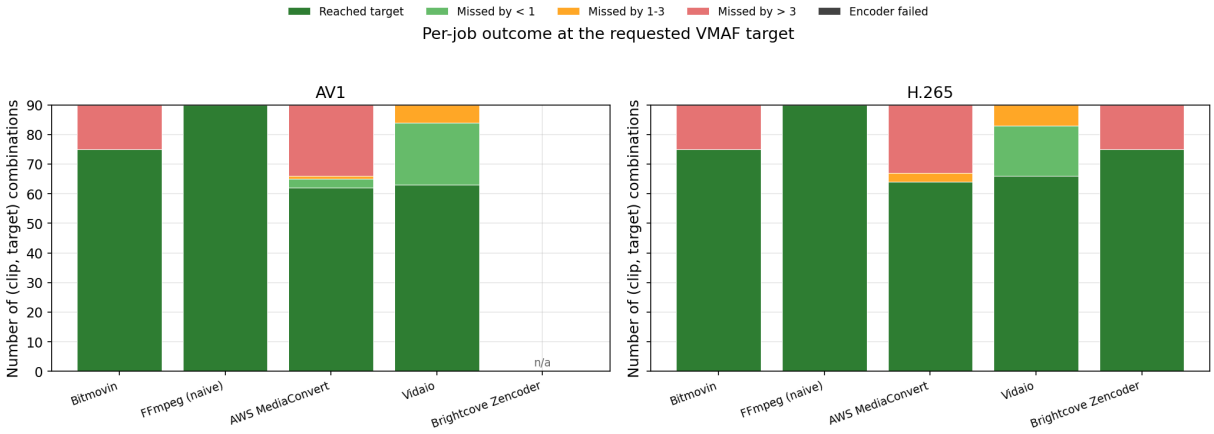
Figure 4.2.6 - RD Curves, H.265, Security & Surveillance Clips



Low-motion surveillance content compresses tightly and the encoders cluster closely.

4.3 Target Attainment

Figure 4.3 - Per-job outcome at the requested VMAF target



Stacked bar per provider × codec across the full test matrix (30 clips × 3 VMAF targets = 90 jobs per provider). Dark green = reached target. Light green = missed by less than 1 VMAF point (effectively at target). Amber = missed by 1-3 points. Red = missed by more than 3 points (encoder hit a ceiling on this content). Dark gray = encoder failed to produce any output.

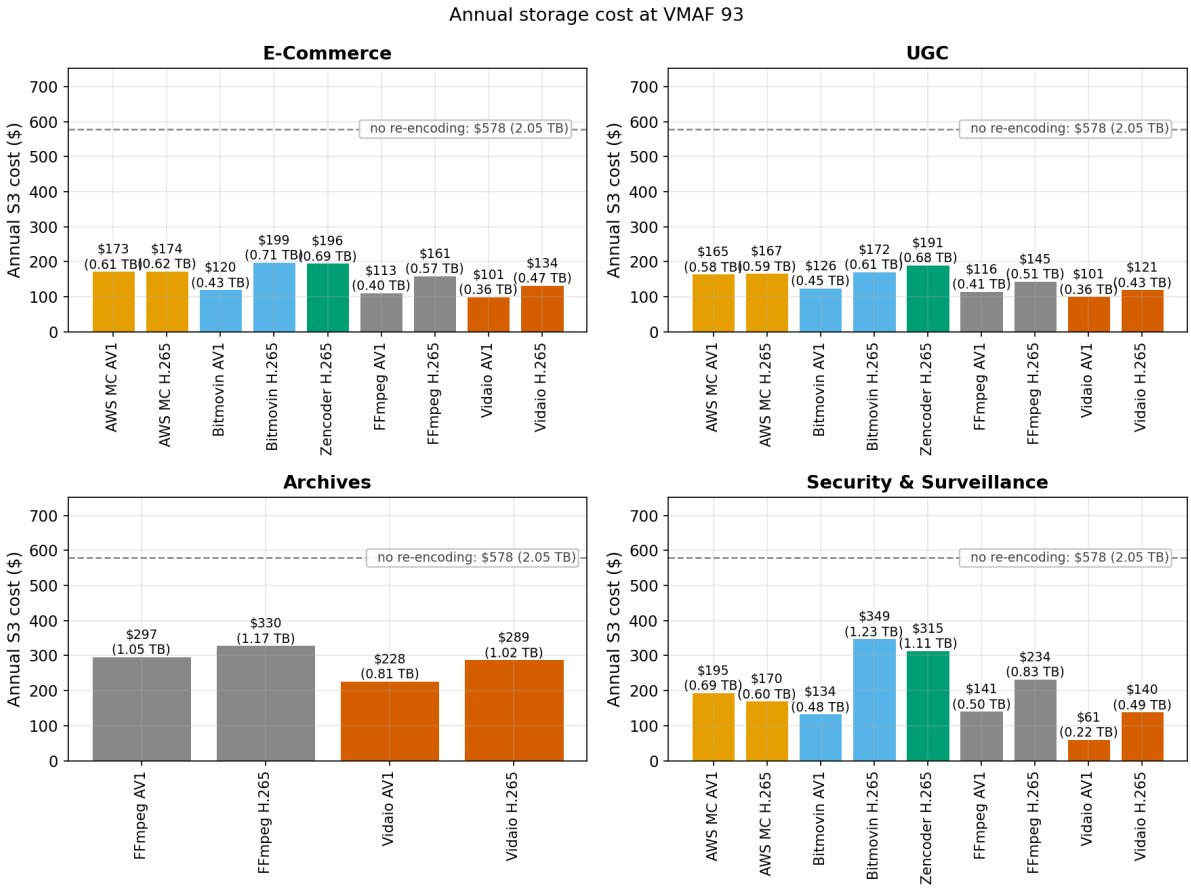
5. Storage Efficiency Analysis

This section translates rate-distortion data into the metrics that storage-cost buyers care about.

Storage cost projection

What it would cost to store 1,000 hours of representative 1080p source (5000 kbps) on S3 ($0.023/GB/mo) for one year, after encoding through each provider at VMAF 93, broken down by vertical. The dashed line shows what storing the raw source without re-encoding would cost.

Figure 5.1 - Annual Storage Cost per 1000 hours of video



Lower delivered bitrate means lower storage cost, so the ranking tracks compression efficiency. Providers that could not reach VMAF 93 on a vertical — the managed encoders on archive — have no bar there.

6. Pricing Analysis

Per-encode price (1080p, public list)

What each provider charges per minute of encoded output.

Provider	H.265	AV1	Pricing Model	Notes
AWS MediaConvert	$0.048	$0.084	Tiered, base × codec × resolution × frame-rate multiplier	Pro Tier 1 base $0.012/ normalized-min × multipliers; volume tier discounts at 100K/1M/10M min/region/mo
Bitmovin	$0.100	$0.500	Pay-as-you-go: base × resolution × codec × pass count	2-pass default; 2000 free min/mo on VOD; AV1 is a 5× surcharge vs HEVC
Brightcove Zencoder	$0.050	—	Pay-as-you-go ~$0.05/min; best tier $2k/mo for 100K min ($0.020/ min)	AV1 not supported; volume-tier discounts via subscription plans
FFmpeg (naive)	-	-	Self-hosted (your hardware/ electricity)	Open-source libx265 / libsvtav1 driven by an interpolation search over CRF only; no list price - cost depends on your infrastructure
Vidaio	$0.067	$0.067	Credit bundles, flat rate (no codec/resolution split)	$20 bundle = 300 min compression at $0.067/min; smallest $5 bundle is $0.083/ min

Per-target price

What it actually costs to deliver one minute of source video at a target VMAF. Vidaio targets VMAF directly with a single encode; the others use an iterative search and every pass is billed. Per-target price = per-encode price × average passes to reach the target.

Provider	Avg passes (H.265)	Avg passes (AV1)	Per-target $/min (H.265)	Per-target $/min (AV1)
AWS MediaConvert	4.0	4.1	$0.193	$0.345
Bitmovin	3.7	4.6	$0.374	$2.289
Brightcove Zencoder	2.8	-	$0.141	-
FFmpeg (naive)	4.0	4.2	-	-
Vidaio	1.0	1.0	$0.067	$0.067

Assumes iterative VMAF search; not list pricing.

Pricing caveat: All costs use public list / pay-as-you-go pricing as of 2026-06-17. Enterprise/volume pricing varies - Brightcove Zencoder's best subscription tier is $0.020/min (100K min/mo plan), and AWS MediaConvert offers volume discounts above 100K min/region/mo. AWS MediaConvert prices are computed from the published multiplier table × Professional Tier 1 base rate.

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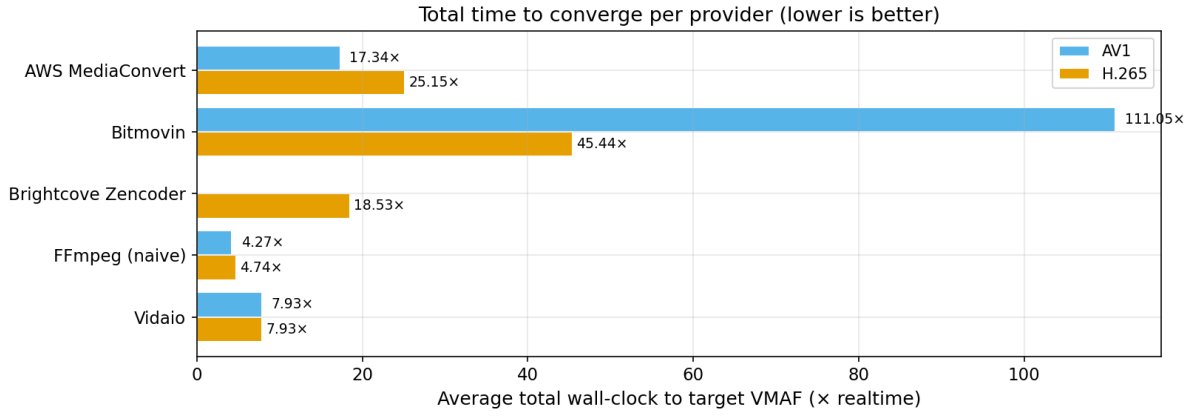
7. Encoding Speed

Wall-clock time from job submission to a passed result, expressed as **× realtime** (compute-seconds / source-seconds). Values below 1.0 mean the encoder finishes faster than the source plays back; values above 1.0 mean it takes longer than the source duration. Per-clip values are normalised by source duration so 1080p (≈120s) and 4K (10s) clips compare directly, then averaged across the corpus. Vidaio's per-encode wall-clock is roughly flat at ≈200s (overhead-dominated); applied per clip its xRT reflects the duration mix.

All speed values are in **× realtime** (compute-seconds / source-seconds).

Provider	Codec	Per-pass (×rt)	Passes	Total to target (×rt)
AWS MediaConvert	AV1	4.87	4.0	17.34
AWS MediaConvert	H265	6.16	3.9	25.15
Bitmovin	AV1	27.63	4.6	111.05
Bitmovin	H265	12.76	3.8	45.44
Brightcove Zencoder	H265	6.79	2.8	18.53
FFmpeg (naive)	AV1	1.01	4.1	4.27
FFmpeg (naive)	H265	1.18	4.0	4.74
Vidaio	H265	7.93	1.0	7.93
Vidaio	AV1	7.93	1.0	7.93

The **Passes** column is the per-target encode budget from §3.2: Vidaio runs a single encode while the others run an iterative search, so even where per-pass speed is comparable, the search multiplies the others' total time to reach a target.

> **Figure 7.1 - Total wall-clock to target VMAF per provider**
>
> 
>
>
> *Total compute time to deliver one converged result, normalised by source duration and averaged across the corpus. Lower is faster; below 1.0× means sub-realtime.*

This is end-to-end wall-clock - it includes provider queue time, file upload, encode, download, and our local VMAF measurement between iterations. FFmpeg has none of the network or queue overhead since it runs locally on a 64-core server with 24-way scene-parallel encoding, so its number reflects what an operator with dedicated hardware would actually see. It is *not* an isolated

encoder-speed measurement; that's not observable from outside a cloud API. Numbers are directional and will vary with region, load, and account tier.

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8. Limitations and Future Work

Limitations

1. **Sample size.** 30 clips is representative but not exhaustive. Results may vary on content types not tested (e.g., sports, gaming, medical imaging).
2. **Provider evolution.** Cloud encoders are updated frequently. These results reflect provider capabilities as of 2026-06-17.
3. **Pricing not fully compared.** Encoding, storage, and delivery costs vary by volume, contract, and region. Storage projections use public list prices and do not account for negotiated enterprise rates.
4. **Encoding time variability.** Cloud encoding speed depends on server load and is not deterministic.

Future work

- Test VP9 and H.264 as additional codec baselines
- Re-run annually as providers update their encoders
- Include cost-per-quality-point analysis with actual invoiced costs

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9. Conclusions

Across 22 clips and three VMAF targets, no single encoder wins everything. Each option has a clear strength and a clear cost.

Vidaio is the most compression-efficient encoder tested. It produces smaller files than every other option at matched VMAF — about 10% fewer bits than the FFmpeg open-source baseline on H.265, 9% fewer on AV1, and 15-43% fewer than the managed cloud providers (Bitmovin, AWS MediaConvert, Brightcove Zencoder). It is the only managed service in this benchmark that beats the open-source reference on compression. It hits target strictly on 68% of jobs, within 1 VMAF point on 95% of jobs, and within 1.5 VMAF points on 100% of jobs.

FFmpeg matches Vidaio on reliability but not on operational profile. FFmpeg strictly meets target on 100% of the matrix and handles content (e.g. grainy archive footage) that every cloud provider fails on. Achieving this requires running, scaling, and monitoring encoding infrastructure yourself, and wrapping the encoder with an iterative search loop to hit VMAF targets — the same

kind of search loop that multiplies cloud providers' per-encode costs. Vidaio reaches comparable compression through a single API call with no iterative search and no infrastructure to operate. For teams with existing video-engineering capacity, self-hosted FFmpeg can be cheaper in absolute terms; for everyone else, Vidaio is the closest approximation of its efficiency through a managed interface.

The cloud providers (Bitmovin, AWS MediaConvert, Brightcove Zencoder) lose on both axes that matter for a managed encoder. They use 14-21% more bits than FFmpeg on H.265 and 7-59% more on AV1, while charging 2-36× more per encoded minute than Vidaio. None of them handles archive content. Their value proposition is managed-API convenience and existing enterprise relationships, but Vidaio delivers the same managed model with materially better compression and lower cost.

Cost differences compound at scale. Per-target price - what you actually pay to deliver one minute of source video at a target VMAF - ranges from $0.067/min (Vidaio, flat-rate, single pass) to $2.42/min (Bitmovin AV1, where the per-encode AV1 surcharge is 5× HEVC and the iterative search averages 4.8 passes). For a workload of 1,000 hours per year, that's the difference between roughly $4,000 and $145,000 in encoding fees alone, before storage. The published per-encode price is misleading because providers without native VMAF targeting need three to five search iterations to converge, each billed. FFmpeg has no list price; its real cost is the hardware and operations to host it.

Appendices

Appendix A: BD-Rate Computation Methodology

Definition: Bjontegaard Delta Rate (BD-Rate) measures the average percentage difference in bitrate between two rate-distortion curves at equivalent quality levels (Bjontegaard, 2001).

Curve fitting: Piecewise cubic Hermite interpolation (PCHIP) of log-bitrate as a function of VMAF.

Integration range: Limited to the VMAF range where both providers have data points, to avoid extrapolation artifacts.

Edge cases: Black-box providers with a single data point are excluded from BD-Rate computation.

Appendix B: Test Content Source Details

Clip	License	Source URL	Resolution	Frames	Pixel Format
archive_achievement_usa_1955	Public Domain	https://archive.org/download/Achievem1955/Achievem1955.ogv	400x300	3596	yuv420p
archive_coke_1954	Public Domain	https://archive.org/download/Coke_Commercial/Coke_Commercial.ogv	400x300	1697	yuv420p
archive_desoto_1955	Public Domain	https://archive.org/download/DeSoto5/DeSoto5.ogv	400x300	2457	yuv420p
archive_experiments_1940	Public Domain	https://archive.org/download/Experime1940/Experime1940.ogv	400x300	3596	yuv420p
archive_spot_news_1937	Public Domain	https://archive.org/download/SpotNews1937/SpotNews1937.ogv	400x300	3596	yuv420p

Clip	License	Source URL	Resolution	Frames	Pixel Format
ecom_iphone 17	Amazon	https://m.media-amazon.com/images/S/vse-vms-transcoding-artifact-eu-west-1-prod/96171188-9a45-4cfd-a6e5-d508d1295a3a/default.jobtemplate.hls.m3u8	1920x1080	1375	yuv420p
ecom_mountain_bike	Amazon	https://m.media-amazon.com/images/S/vse-vms-transcoding-artifact-eu-west-1-prod/12f1ae85-6ea2-42d7-a36b-97a30c19db92/default.jobtemplate.hls720.m3u8	1280x720	3597	yuv420p
ecom_nike_sneakers	Amazon	https://m.media-amazon.com/images/S/vse-vms-transcoding-artifact-eu-west-1-prod/6db80be0-ff94-435b-8f34-f5b137efe2c6/default.vertical.jobtemplate.hls.m3u8	1080x1440	450	yuv420p

Clip	License	Source URL	Resolution	Frames	Pixel Format
ecom_supplements	Amazon	https://m.media-amazon.com/images/S/vse-vms-transcoding-artifact-eu-west-1-prod/f04ad07d-98e3-49fa-9a64-b4baf3094a46/default.jobtemplate.hls720.m3u8	1280x720	744	yuv420p
ecom_watch	Amazon	https://m.media-amazon.com/images/S/vse-vms-transcoding-artifact-eu-west-1-prod/f096a067-6581-4456-b776-f5746f5f76bb/default.jobtemplate.hls.m3u8	1920x1080	307	yuv420p
security_virat_001	VIRAT Dataset Agreement	(restricted dataset)	1920x1080	3597	yuv420p
security_virat_002	VIRAT Dataset Agreement	(restricted dataset)	1920x1080	3597	yuv420p
security_virat_003	VIRAT Dataset Agreement	(restricted dataset)	1920x1080	3597	yuv420p
security_virat_004	VIRAT Dataset Agreement	(restricted dataset)	1920x1080	3597	yuv420p
security_virat_006	VIRAT Dataset Agreement	(restricted dataset)	1920x1080	3597	yuv420p
security_virat_101	VIRAT Dataset Agreement	(restricted dataset)	1920x1080	3597	yuv420p

Clip	License	Source URL	Resolution	Frames	Pixel Format
security_virat_102	VIRAT Dataset Agreement	(restricted dataset)	1920x1080	3597	yuv420p
ugc_4k_battlefield_gameplay	YouTube	https://www.youtube.com/watch?v=ZZYuIgDB32o	3840x2160	7800	yuv420p10le
ugc_4k_gym_motivation	YouTube	https://www.youtube.com/watch?v=mpL7e0_jLXQ	3840x2160	3897	yuv420p
ugc_4k_kpop_dance	YouTube	https://www.youtube.com/watch?v=qORaYudQ7Zc	3840x2160	7793	yuv420p
ugc_4k_monaco_walk	YouTube	https://www.youtube.com/watch?v=oS-rwVafano	3840x2160	7793	yuv420p10le
ugc_4k_pasta_asmr	YouTube	https://www.youtube.com/watch?v=IkalikR-pFs	3840x2160	7800	yuv420p
ugc_4k_skating_fpv	YouTube	https://www.youtube.com/watch?v=vSBcrmx4aFw	3840x2160	3117	yuv420p
ugc_4k_switzerland_drone	YouTube	https://www.youtube.com/watch?v=LQuLAbG62vY	3840x2160	3897	yuv420p

Clip	License	Source URL	Resolution	Frames	Pixel Format
ugc_4k_vienna_shortfilm	YouTube	https://www.youtube.com/watch?v=xck6vJpmevY	3840x2160	3250	yuv420p
ugc_cooking_ramsay	YouTube	https://www.youtube.com/watch?v=mzDCuS2l_ql	1920x1080	21237	yuv420p
ugc_guitar_performance	YouTube	https://www.youtube.com/watch?v=8UPHFjHvGvY	1280x720	6174	yuv420p
ugc_laptop_unboxing	YouTube	https://www.youtube.com/watch?v=XHFNeEmSjmY	1920x1080	26001	yuv420p
ugc_skate_tricks	YouTube	https://www.youtube.com/watch?v=sjAt8m1Vvyw	1920x1080	38323	yuv420p
ugc_vlog_finland	YouTube	https://www.youtube.com/watch?v=2eloJfB7gRY	1920x1080	31658	yuv420p

Appendix C: Glossary

Term	Definition
P-frame	Predicted frame. References previous frames to achieve higher compression.
B-frame	Bidirectionally predicted frame. References both previous and future frames for maximum compression.
RD curve	Rate-Distortion curve. Plots quality (VMAF) against bitrate. A curve closer to the top-left is more efficient.
SI	Spatial Information. Measures the spatial complexity (detail) of a video frame. Higher SI = more detail.
TI	Temporal Information. Measures the temporal complexity (motion) between frames. Higher TI = more motion.

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